

Debdip Bandyopadhyay

Lead SDET & AI Quality Engineering Lead

Kolkata, India | +91-8285578544 | debdip1992@outlook.com | linkedin.com/in/debdiparvr
github.com/debdipARVR | huggingface.co/spaces/DebdipCS/forensics-hub

PROFESSIONAL SUMMARY

Lead Software Development Engineer in Test (SDET) and AI Quality Engineer with **11+ years** of experience architecting enterprise test frameworks, non-deterministic evaluation harnesses, and CI/CD quality gates across Generative AI platforms, Aerospace, and Enterprise Systems. Core quality lead in the **GenAI Center of Excellence (CoE)** at Cognizant, validating production LLM/RAG pipelines using DeepEval, semantic similarity, and custom automated metrics. Holds an **M.Tech from IIT Jodhpur (School of AI & Data Science)** with independent published research and CERN Zenodo DOIs in foundation model provenance traceback, black-box AI text forensics, and autoencoder latent invariants.

TECHNICAL SKILLS

AI Quality & LLM Evaluation: LLM Testing, RAG Triad Benchmarking (Faithfulness, Answer Relevancy, Context Recall), DeepEval, Hallucination Scoring, Semantic Similarity, Prompt Injection Testing, Non-Deterministic Output Validation, PyTorch.

Automation Frameworks: Python, Playwright, Pytest, Pytest-BDD, Cypress, Page Object Model (POM), Parallel Execution (pytest-xdist), REST API Automation (Requests, Postman v12), Selenium WebDriver.

CI/CD & Cloud Infrastructure: Azure DevOps Pipelines, Continuous AI Evaluation Gates, Docker Containerization, Poetry, Git, Linux / Shell Scripting.

Enterprise Systems & Verification: Oracle EBS (AP, AR, PO, CE, FA), Tricentis Tosca, UFT, Zero-Trust Quality Gates, Jira, Agile/Scrum.

PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions

Kolkata, India

Official Corporate Designation: Developer / Project Associate

Apr 2021 – Present

Lead AI Quality Engineer / Senior SDET – GenAI CoE

Sep 2025 – Present

- Serve as technical quality lead across cross-domain squads (NLP researchers, full-stack engineers, cloud architects) establishing zero-trust evaluation gates for production LLM and RAG conversational applications.
- Architected automated CI/CD evaluation pipelines using **DeepEval**, validating non-deterministic outputs across faithfulness, correctness, context relevancy, and semantic similarity to prevent hallucination drift before release.
- Designed and scaled an enterprise-grade **Playwright + Python + Pytest-BDD** automation framework utilizing modular Page Object Model (POM) patterns and custom API fixtures.
- Reduced enterprise regression cycle times by **~40%** through selective test sharding, headless browser optimization, and distributed multi-worker execution via **pytest-xdist**.
- Embedded automated quality gates into **Azure DevOps Pipelines**, blocking pull requests when hallucination rates or context-drift deltas breach production thresholds.
- Mentored 10+ automation engineers on Python test patterns and aligned enterprise quality benchmarks with Solution Architects and Data Scientists.

Senior Test Automation Engineer – Enterprise Systems (Oracle QEA)

Apr 2021 – Aug 2025

- Architected modular automation frameworks using UFT and Tricentis Tosca supporting multi-module, multi-Operating Unit (OU) execution across Oracle EBS suites (AP, PO, FA, CE, AR).
- Engineered custom framework utilities featuring runtime state persistence, resume-from-failure logic, automated credential handling, and intelligent synthetic test data generation.
- Delivered client PoCs and technical demonstrations for Fortune 500 hospitality conglomerates and global financial services accounts across APAC, EMEA, and North America.
- Automated high-volume regression suites across 500+ critical financial workflows during enterprise CPU patch release cycles.

The Boeing Company

Bengaluru, India

Software Engineer – Test Automation

Oct 2018 – Apr 2021

- Engineered test automation solutions for safety-critical aerospace software adhering to stringent aerospace engineering regulations and compliance standards.
- Maintained and executed enterprise regression suites exceeding **2,000+ automated test cases** across multi-phase release cycles.
- Enforced automated test coding standards, CI integration checks, and technical code reviews across junior engineering teams.

- Recipient of the **Boeing Pride Award** (safety-critical flight platform validation) and **Star Award** (2019/2020) for outstanding sprint delivery and test optimization.

Accenture LLP

Kolkata / Noida / Bengaluru, India

Test Automation Engineer

Oct 2014 – Oct 2018

- Built reusable, data-driven automation frameworks across web, client-server desktop, SAP MM, and mobile environments for telecommunications and energy sector clients.
- Automated end-to-end integration, system regression, smoke, and User Acceptance Testing (UAT) sign-off workflows.
- Designed and deployed an automated enterprise test-capture utility that reduced third-party tooling licensing expenses by ~\$8,000 annually.

PUBLICATIONS & AI RESEARCH PREPRINTS

Genre-Routed Generative Resonance: Black-Box AI Text Detection via Dual-Horizon Probing and Structural Rhythm

SSRN (Sep 2026) – Logit-free forensic framework solving the Reasoning Prober Paradox and ESL bias via genre routing. Evaluated on $N = 1,000$ documents across frontier models (GPT-4o, Claude 3.5, LLaMA-3.3); achieved **0.8283 AUROC** with **0.00% ESL False Accusation Rate** at disciplinary threshold ($\tau \geq 0.38$).

ClozeCongruence: Model Fingerprinting and Traceback Attribution in LLMs via Multi-Density Reverse Cloze Resonance

CERN Zenodo [DOIs: [10.5281/zenodo.22158286](https://doi.org/10.5281/zenodo.22158286), [10.5281/zenodo.22158483](https://doi.org/10.5281/zenodo.22158483)] – Formulates 12D micro-stylometric analysis and reverse cloze probing to isolate latent model DNA in black-box environments; achieved high Top-1 attribution accuracy across frontier architectures.

Latent Resonance: Zero-Shot Autoencoder Inversion and Azimuthal Spectral Forensics for Diffusion Image Attribution

SSRN (Sep 2026) – Isolates spatial manifold resonance and transposed-convolution harmonic spikes via deterministic autoencoder projection at $\sigma = 0$; achieved **72.00% AUROC** with resilience under lossy JPEG compression and Gaussian blur.

The Cloze Scaling Laws: Prober Parameter Dynamics, Native Blank-Infilling Architecture, and Traceback

CERN Zenodo / SSRN 7368141 (2026) – Empirically maps how prober parameter scale (12B to 120B) and native masking architectures govern attribution recovery bounds in generated discourse.

EDUCATION

Indian Institute of Technology Jodhpur (IIT Jodhpur)

Jodhpur, India

Master of Technology (M.Tech) – Augmented Reality & Virtual Reality

Jul 2022 – Dec 2025

- **Department:** School of Artificial Intelligence & Data Science (AIDE).
- **Focus:** Deep Learning, Advanced Computer Vision, Generative Modeling, Latent Space Forensics, Spatial Systems.

Greater Noida Institute of Technology (AKTU / UPTU)

Greater Noida, India

Bachelor of Technology (B.Tech) – Computer Science & Engineering

Apr 2010 – Jun 2014

Kendriya Vidyalaya No. 1, Salt Lake

Kolkata, India

Higher Secondary Certificate (CBSE Class XII) – PCMB

Completed: 2010

- **Honors:** State Rank Top 100 – Science Olympiad (Class 11).

CERTIFICATIONS & HONORS

Certifications: Postman API Testing Path (v12) | ASDA-MIT Application Tester | Google Crash Course on Python (Coursera).

Corporate Accolades: Boeing Pride Award (Aerospace System Validation) | Boeing Star Award | Accenture Asset Creation Award.